C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 6.01

Permutations & Combinations

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Permutations & Combinations Practice Exercises

1 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$$

A $n = 7$

B $n = 12$

C $n = 5$

D $n = 8$

2 MCQ

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

A 13

B 10

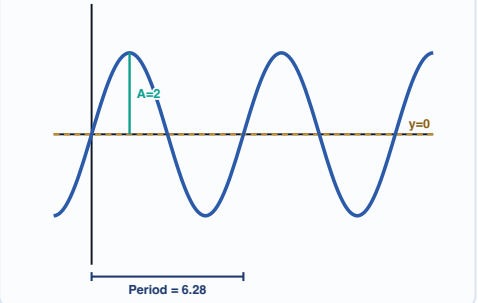
C 8

D 7

3 MCQ

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number.

Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.



A 8160

B 7680

C 9120

D 2880

4 MCQ

Given that

$$^{n}\mathrm{P}_{3} = 8\,^{n}\mathrm{C}_{2},$$

find the value of the integer n, where $n \geq 3$.

A 6 B 8

C 10 D 12

5 MCQ

A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600. Find the value of the integer n.

A 6 B 7

C 8 D 9

6 MCQ

Two parallel lines, L_1 and L_2, have n and $n+2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$.
Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear.
Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.

A 175 B 210

C 220 D 96

7 MCQ

Given that $^{n}\mathrm{P}_{2} = \binom{n}{3}$, where n is an integer such that $n \geq 3$, find the value of n.

A 8 B 5

C 3 D 6

8 MCQ

A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

A 60 B 120

C 55 D 36

9 MCQ

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 4)$. What is the value of the constant a?

A -10.00

B 7.00

C -7.00

D -5.00

10 MCQ

Work out the solution to the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{6}{\ln 5 - \ln 2}$

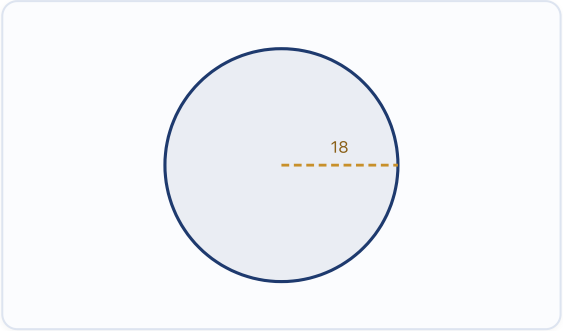
B $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

C $x = \frac{6 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{6 \ln 2}{\ln 5 - \ln 2}$

11 MCQ

Consider a sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?



A $453.6\ \text{cm}^2$

B $226.8\ \text{cm}^2$

C $25.2\ \text{cm}^2$

D $231.8\ \text{cm}^2$

12 MCQ

A committee of 4 people is to be chosen from 6 men and 5 women. How many committees consist of exactly 2 men and 2 women?

A 44

B 150

C 145

D 300

13 MCQ

In an applied problem, a committee of 4 people is to be chosen from 7 men and 6 women. How many committees consist of exactly 2 men and 2 women?

A 310

B 630

C 315

D 52

14 MCQ

Suppose a committee of 4 people is to be chosen from 8 men and 7 women. How many committees consist of exactly 2 men and 2 women?

A 588

B 583

C 1176

D 60

15 MCQ

An observed committee of 4 people is to be chosen from 9 men and 8 women. How many committees consist of exactly 2 men and 2 women?

A 1008

B 68

C 2016

D 1003

16 MCQ

Consider a committee of 4 people is to be chosen from 10 men and 9 women. How many committees consist of exactly 2 men and 2 women?

A 1615

B 1620

C 3240

D 76

17 MCQ

A committee of 4 people is to be chosen from 11 men and 10 women. How many committees consist of exactly 2 men and 2 women?

A 2475

B 2470

C 4950

D 84

18 MCQ

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

A $\frac{80}{2}$

B $\frac{84}{8}$

C $\frac{80}{8}$

D $\frac{80}{4}$

19 MCQ

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-3} \quad (x \neq 3)$

B $f^{-1}(x) = \frac{3x-2}{x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-3}$

D $f^{-1}(x) = \frac{x-2}{3x+1}$

20 MCQ

Considering the expansion of $(1+\frac{x}{3})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

A 12
B 13
C 7
D 15

21 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x)-\log_2(y)=2$ and $x-3y=5$. What is the value of x?

A 5
B 22
C 10
D 20

22 MCQ

Find all solutions of $2\sin^2 x-3\sin x+1=0$ in the interval $0\le x\le 2\pi$.

A $x=\frac{\pi}{3},\ \frac{2\pi}{3}$
B $x=\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$
C $x=\frac{3\pi}{2}$
D $x=\frac{7\pi}{6},\ \frac{11\pi}{6}$

23 MCQ

Find the number of distinct permutations of all letters in PARALLEL if all 3 'L's must stay together as a single block.

A 180
B 720
C 3360
D 360

24 MCQ

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

A 1680
B 3360
C 6720
D 50400

25 MCQ

Find the number of distinct permutations of all letters in COMMITTEE if the 2 'M's must stay together.

A 20160
B 45360
C 10080
D 5040